Design of Bifunctional Fentanyl Analogs: Dual-Targeting the Mu-Opioid and Nociceptin Receptors for Safer Analgesia

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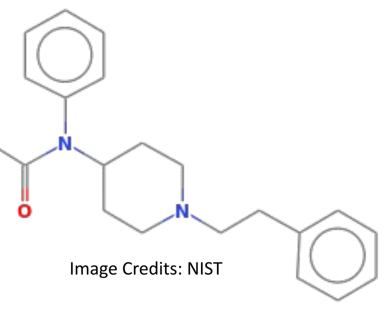
Introduction

Fentanyl is the primary driver of opioid overdose. Between 2019 and 2023, Oregon experienced a 1,500% increase in fentanyl-related overdose deaths, the highest in the United States.

Fentanyl Overview

Low-cost synthetic **opioid** that is **100x** stronger than heroin & morphine. Widely used for its **potency** and **fast-acting** analgesic effects. Binds to the Mu-Opioid Receptor (MOR) in the Central Nervous System via the **Asp147** residue. [2]



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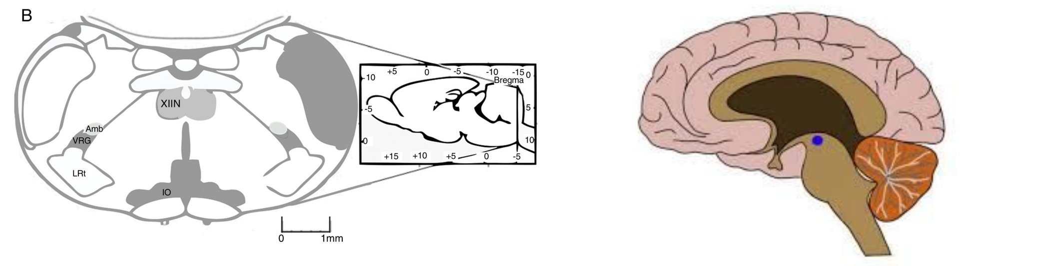
Challenges



PreBotzinger Complex [1] Ventral Tegmental Area [2]

- Binding to the MOR in the PreBötzinger Complex induces respiratory failure [2].
- Binding to the MOR in the Ventral Tegmental Area (VTA) promotes dopamine release and addictive potential. [3].

Existing Solutions

Alternatives

Morphine/Heroin:
- Long history, widely available
- Slower onset, less potency than fentanyl
- Maintains risks of respiratory depression and dependence

Acetaminophen/Ibuprofen:
- Very safe, Over-the-Counter Accessible
- Significantly weaker analgesic effect

Treatments

Naloxone (Narcan):
- **Reverses** Opioid overdose by **blocking** off Mu-Opioid Receptors
- **Symptomatic** measure - Cannot change inherent risk
- **No effect** on addiction potential - Does not inhibit dopamine

Problem Statement: Fentanyl carries heavy side-effects, while no robust alternatives or treatments exist

Research Approach

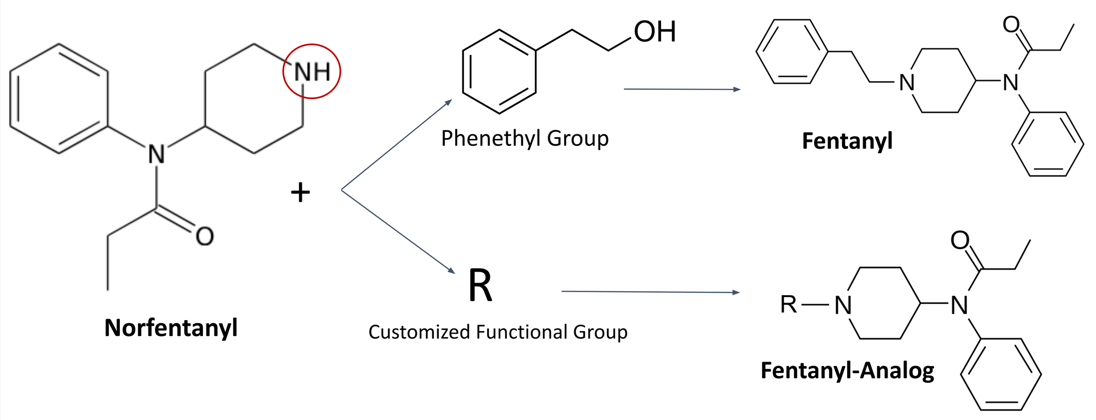
PROBLEM		SOLUTION
MOR Targeting Compounds carry heavy side-effects	—	Nociceptin/Orphanin FQ receptor (NOP)

Pharmacological studies have shown NOP regulates the same systems that MOR overstimulate. Dual-activation can reduce side-effects.

PROBLEM		SOLUTION
Designing completely new drugs is expensive, time-consuming, risky	—	Modification of fentanyl—Preserves validated scaffold

Modifying the fentanyl structure for boosted NOP activation can preserve its well-known pharmacokinetics and synthesis procedure

Proposed Solution



Functionalization of Norfentanyl w/phenethyl group on piperidine nitrogen produces fentanyl. We hypothesize that functionalizing w/alternate groups on piperidine nitrogen can potentially produce fentanyl-adjacent structures with both MOR and NOP affinity

Analog Identification

Framework - Fragment Building

- **Known NOP Drugs** (Cebranopadol, Ro64-6198, SCH-221510, etc.) → **Fragment Decomposition** (Break into groups) → **Reference Tails** (High Similarity Targets)
- **ZINC-22 Fragment Database** → **Screen Similarity via RDKit** (Vs. Reference Tails) → **Top Ranked Fragment** (High Similarity via) → **Fragment Pool**

Explanation: By analyzing commonalities between known NOP agonists, we can recognize functional groups likely to contribute to NOP activity

Similarity Parameters

3D Shape and Electrostatic Similarity	Physicochemical Property	Maximum Common Subgraph Similarity	Quantum Chemical Descriptors

Similarity Calculations

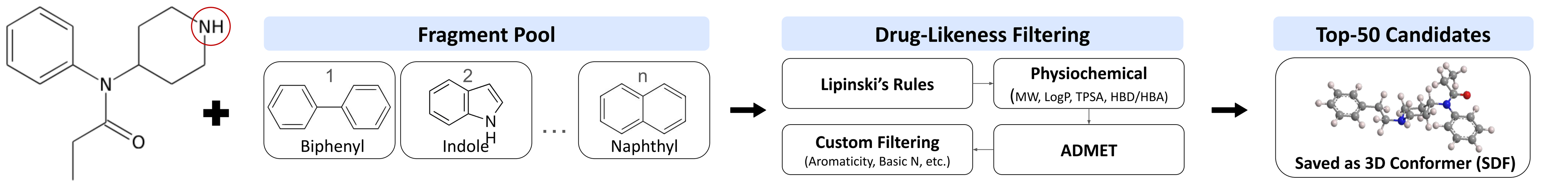
$$S_{Final} = (S_{3D})^{0.40} \times (S_{QM})^{0.30} \times (S_{Phys})^{0.20} \times (S_{MCS})^{0.10}$$

Compounds with a similarity score > 0.8 are saved to fragment pool

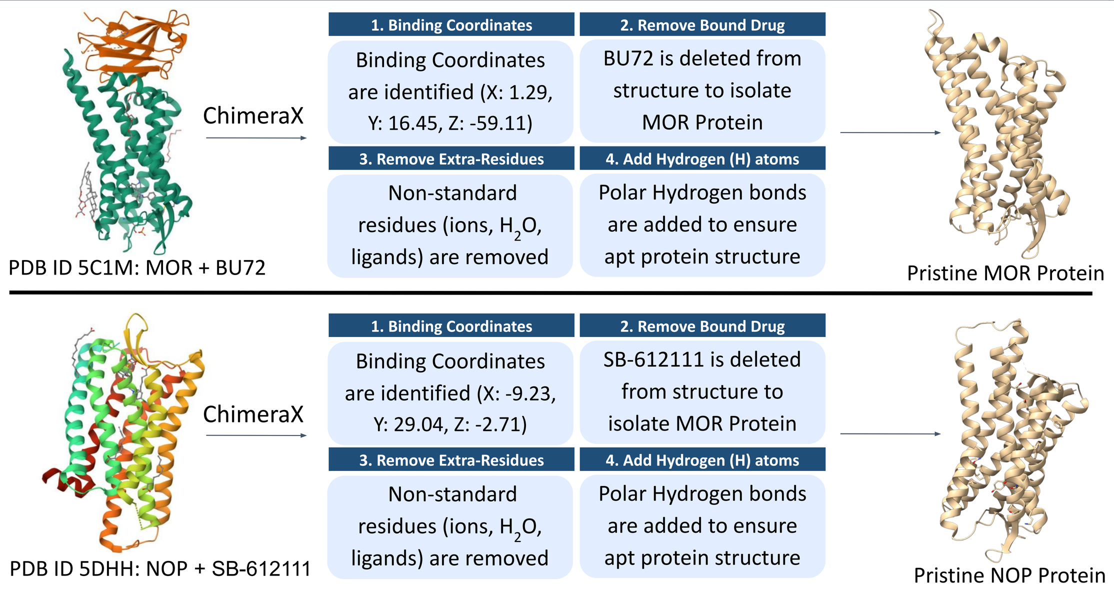
Computational Analog Design

Methods: Analog Generation

In-Silico Preparation

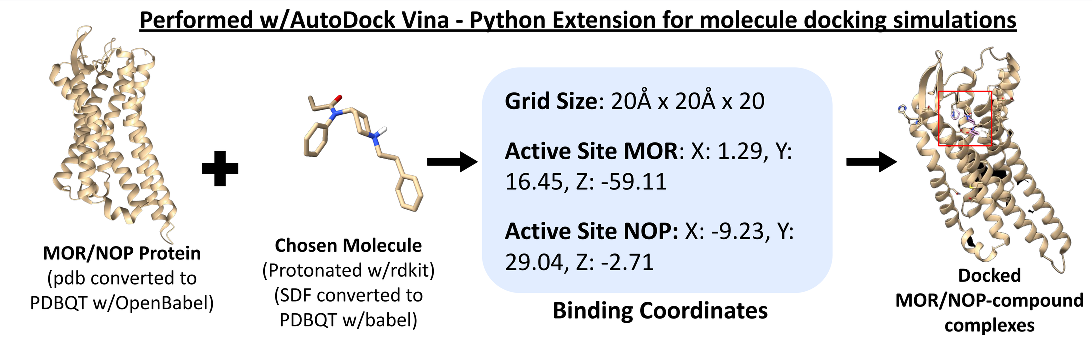


Methods: Target Protein Preparation



Methods: Simulations

Performed w/AutoDock Vina - Python Extension for molecule docking simulations



Vina Scoring Function
Vina calculates the binding affinity of two molecules by estimating gibbs free energy, ΔG, given by:

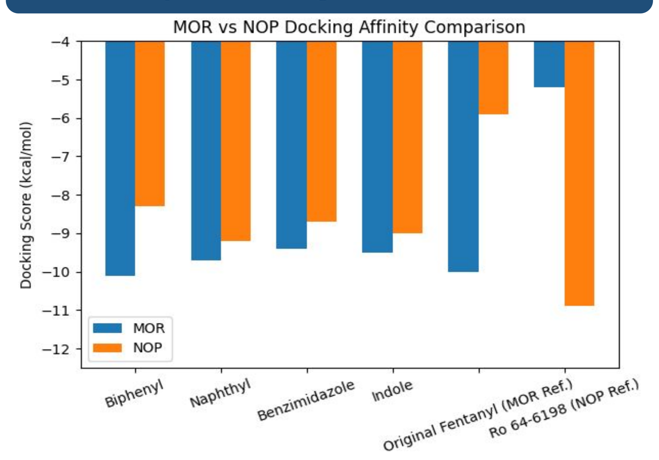
$$\Delta G = \Delta H - T\Delta S$$

(H = Enthalpy, T = Temperature, S = Entropy)
Negative scores indicate High Binding affinity

57 Compounds filtered for MOR & NOP **ΔG<-9.0** kcal/mol and saved for **molecular dynamics** (MD) → 4 resultant compounds. Converted to .xml files. **MD performed** w/OpenMM.

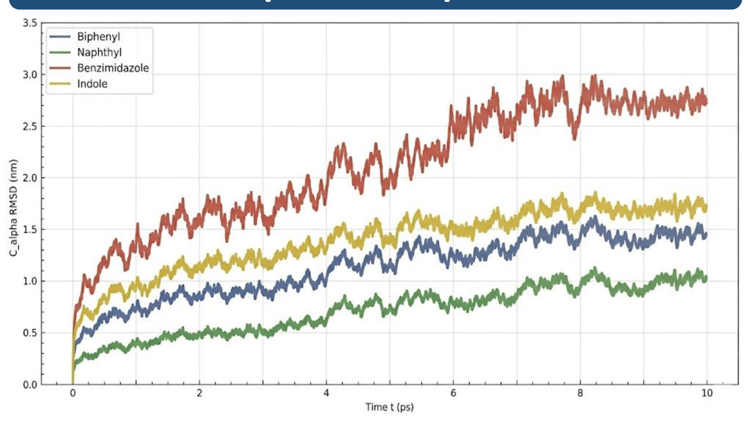
Results: Docking + Molecular Dynamics

Top Docking Candidates

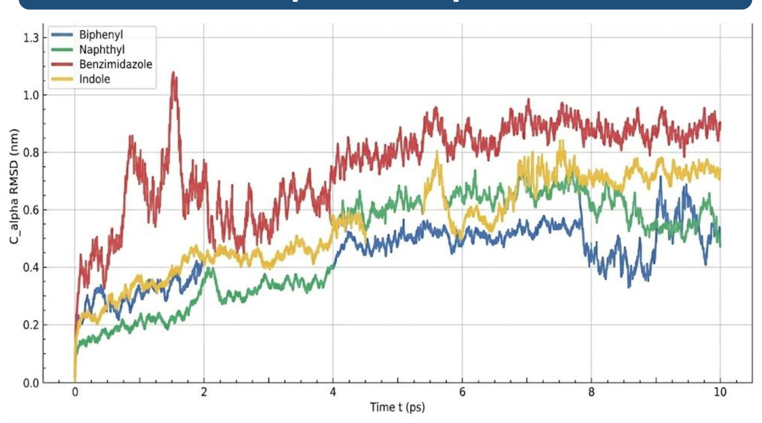


Chosen Functional groups show Comparable MOR/NOP affinity to established agonists

Mu-Opioid Receptor MD



Nociceptin Receptor MD

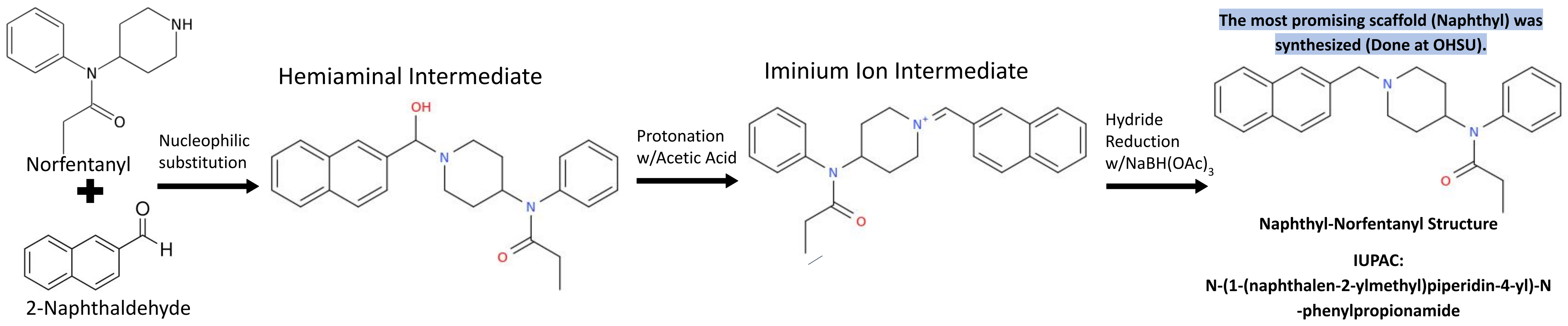


All molecules showed Root Mean Square Deviation (RMSD) of <3.0 Angstrom. Naphthyl modified scaffold displayed lowest RMSD across both MOR and NOP, showing highest promise for in-vitro validation.

Experimental Validation

Methods: Synthesis

One-pot Functionalization of Norfentanyl

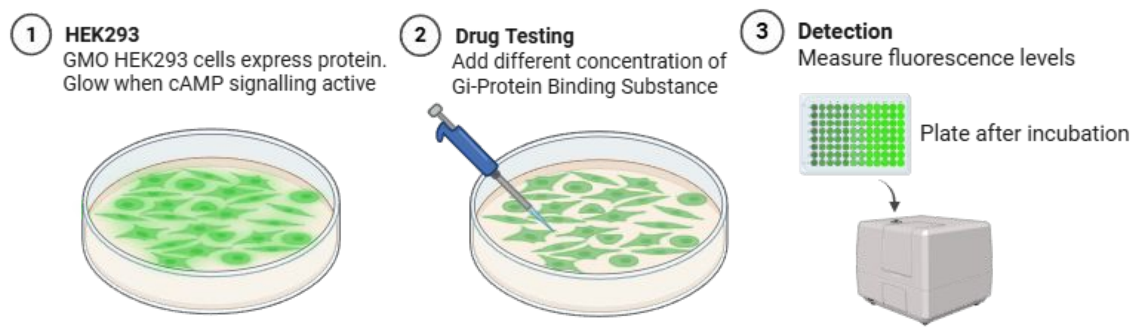


IUPAC:
N-(1-(naphthalen-2-ylmethyl)piperidin-4-yl)-N-phenylpropionamide

Methods: In-Vitro Compound Validation

Individual cAMP Assays

cAMP Assay: a method that measures a cell's signaling activity; cAMP is high (and can produce a fluorescent signal) when Gi-coupled receptors like MOR or NOP are inactive, and drops (dims) when they're activated.



HEK293 Expressing MOR	HEK293 Expressing NOP
cAMP Assay w/4 Candidates (w/Control Fentanyl)	cAMP Assay w/4 Candidates (w/Control n/OFQ peptide)
Extract IC_{50}	Extract IC_{50}

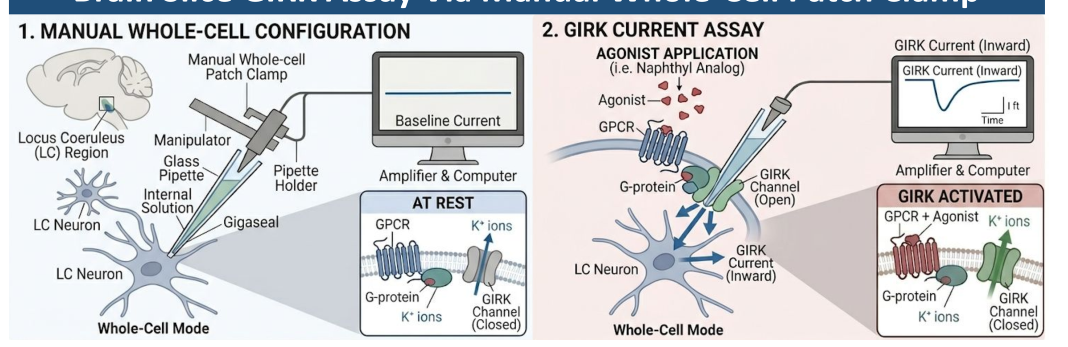
Ideal Result: Balanced Potency at both MOR and NOP

Dual-Expressing cAMP Assay

HEK293 Expressing MOR + NOP
- 4 Opioid Candidates (Test Agonist) → cAMP Suppression (Shallower than MOR-only)
- 4 Candidates + Nop Antagonist (SB-612111) → cAMP Suppression Deepens (NOP Blocked)
- Fentanyl (MOR Only Agonist) → Deep cAMP Suppression (reference)

Electrophysiology

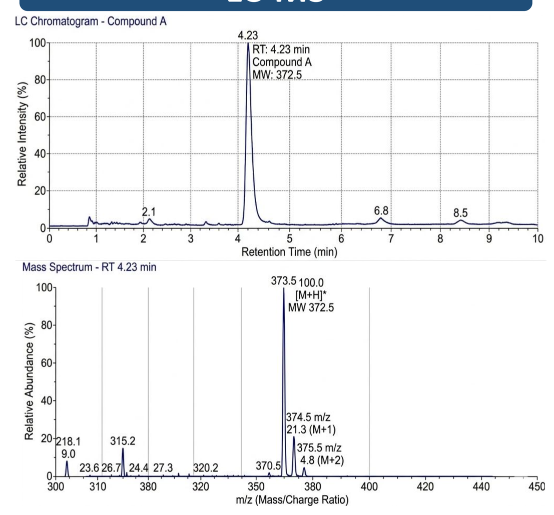
Brain Slice GIRK Assay Via Manual Whole-Cell Patch Clamp



GIRK channels serve as a readout in electrophysiology assays to measure how effectively a drug activates a specific receptor signaling pathway.

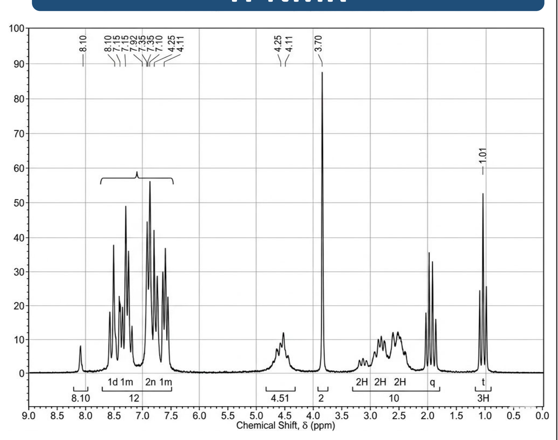
Result: Synthesis Characterization

LC-MS



- LC shows main peak retention time at 4.23 minutes
- MS shows at 4.23 min, primary MW is 372.5 g/mol

H-NMR

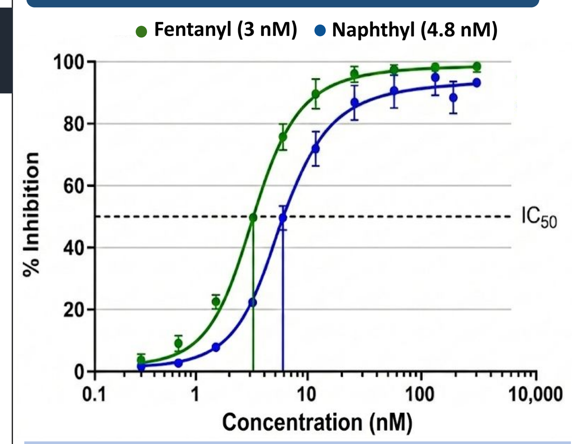


- H-NMR Shows **12 aromatic protons** (naphthyl + phenyl rings), a **2H singlet** at 3.70 ppm marking a **methylene group**, and 2H quartet and 3H triplet between 1.0 and 2.0 ppm showing **propionyl group**

Synthesis is confirmed with ~94% purity via LC-MS and H-NMR

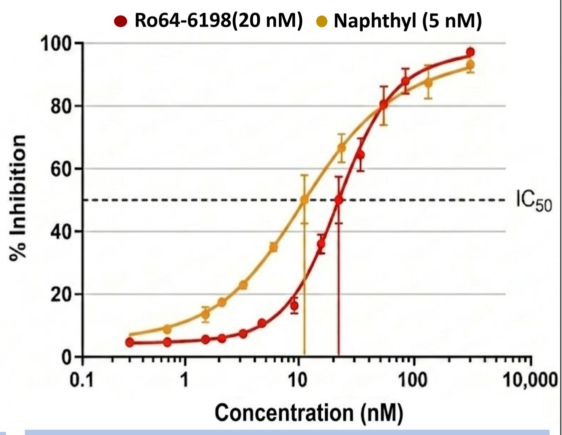
Results: In-Vitro cAMP

IC50 Values - MOR



The Naphthyl compound showed a half-maximal inhibitory value (IC50) of 5 nanomolar. This is comparable to Fentanyl's IC50 of 3 nanomolar, showing high MOR potency

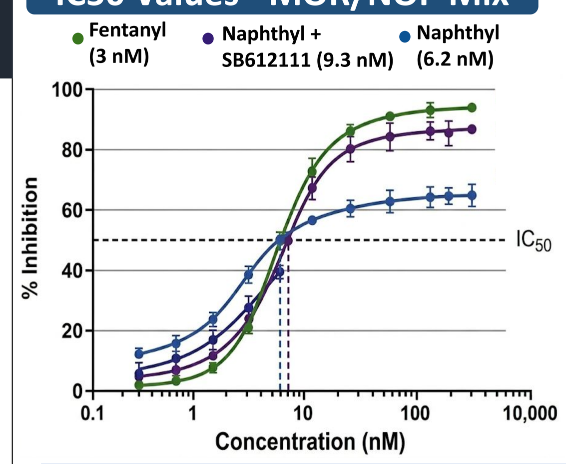
IC50 Values - MOR



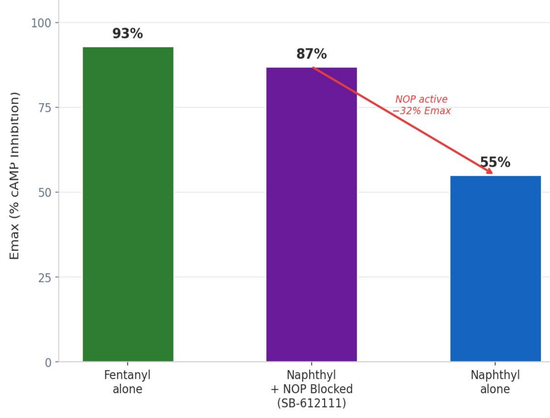
The Naphthyl compound showed an IC50 of 11 nanomolar for the Nociceptin Receptor. This is greater than the Ro64-6198 of 11 nanomoles, showing superior NOP potency

Results: Dual Expressing cAMP

IC50 Values - MOR/NOP Mix



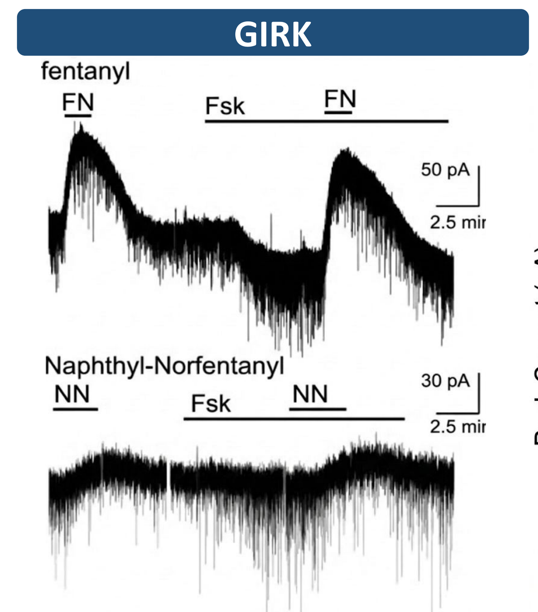
Emax Summary



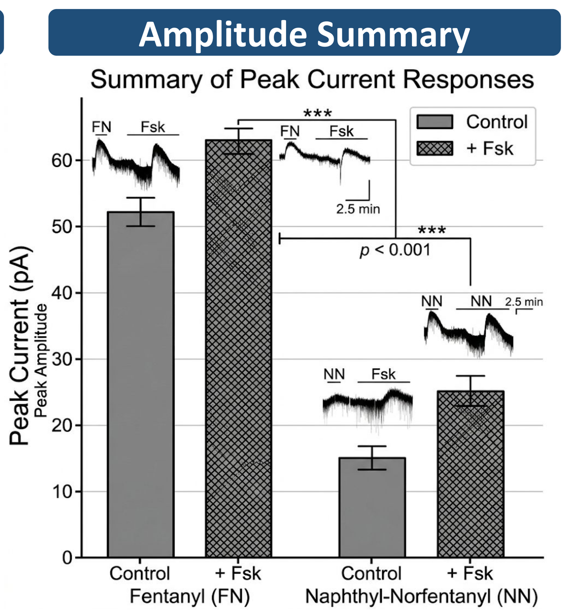
In MOR/NOP co-expressing cells, the Naphthyl compound showed reduced cAMP suppression (55% Emax) compared to fentanyl (93% Emax), demonstrating that NOP activation substantially limits the MOR-driven inhibitory ceiling

Results: Electrophysiology

GIRK



Amplitude Summary



Naphthyl compound showed significantly reduced signaling compared to Fentanyl (30 picoamps vs 50), with peak GIRK currents nearly 2x smaller under control conditions (p < 0.001)

Conclusions

Recap
- **Fentanyl** is a popular potent, fast-acting opioid. Mu-Opioid Receptor binding induces **lethal side-effects** such as **respiratory depression** and **addiction**
- **Identified** and **Synthesized** Novel fentanyl-derivative with **reduced side-effects**

Highlights
- IC50 Values **Comparable** to **Highly-potent independent agonists**
- **Decreased** GIRK Conductivity (responsible for inducing respiratory depression) **by 2x**

Applications
- Enhances pain treatment as a **potent, low-risk alternative** to traditional opioids
- Available as a **safe prescription** w/o monitoring. Useful for **chronic severe pain**
- Helps to **alleviate** opioid overdose crisis

Future Work
- Measure specific binding affinity via radioligand binding assay
- Synthesize and compare the other molecule candidates
- Perform more extensive pharmacokinetic and toxicity testing in-vitro
- Test compounds in-vitro with iPSC-derived brain cells, later in-vivo on mice

Key References

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